Chapter 4. Inference for Two Populations

Rank-sum test, shift estimation, paired comparisons, scale tests, Kolmogorov–Smirnov

Contents

4.1 Models for two populations **2**

4.2 The Wilcoxon rank-sum (Mann–Whitney) test **2**

4.2.1 Large-sample approximation via sampling without replacement 2

4.2.2 Estimating the shift 3

4.3 Tests for scale **3**

4.4 The Kolmogorov–Smirnov test **4**

4.5 Which test? **4**

4.6 Unequal variances under the null **4**

4.7 Selected exercises **5**

4.1 Models for two populations

$X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ independent, $N = m + n$. Typical models:

- **Shift:** $G(x) = F(x - \Delta)$, i.e. $Y \overset{d}{=} X + \Delta$.
- **Scale:** $G(x) = F(x/\gamma)$ (common center 0), or location–scale $G(x) = F((x - \Delta)/\gamma)$.
- **General:** $H_0 : F = G$ against $F \neq G$.

Permutation tests need exchangeability under H_0, i.e. H_0 must force $F = G$ ($\Delta = 0$, $\gamma = 1$, or the general null); "equal locations, different scales" fails (Section 4.6).

For normal data: the pooled t-test (equal variances) or Welch's $T = (\bar{Y} - \bar{X})/\sqrt{s_X^2/m + s_Y^2/n}$ with Satterthwaite d.f.

4.2 The Wilcoxon rank-sum (Mann–Whitney) test

Motivation. Pooled ranks make the permutation distribution data-free, and the rank sum estimates $P(X < Y)$.

Rank all N observations together. Let $R_1, \dots, R_n$ be the ranks of the Y's.

Definition (Rank-sum and Mann–Whitney statistics)

$W = \sum_{j=1}^{n} R_j$ and $U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i)$ (with $\frac{1}{2}$ for ties).

Proposition 4.1

Without ties, $W = U + n(n+1)/2$.

Proof. $R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}$; summing over j the middle terms give $0 + \cdots + (n-1)$, so $W = n + \frac{n(n-1)}{2} + U$. ■

Under $H_0 : F = G$ (continuous) the Y-ranks form a uniform n-subset of $\{1, \dots, N\}$, so the null law of W depends only on (m, n): the permutation test on ranks. Large W indicates $\Delta > 0$.

Example 4.1

$X = \{1.1, 2.4\}$, $Y = \{2.0, 3.5, 4.2\}$: Y-ranks $2, 4, 5$, $W = 11$, $U = 5$. Of the 10 rank subsets only $\{2,4,5\}, \{3,4,5\}$ have sum ≥ 11: $P(W \geq 11) = 0.2$.

4.2.1 Large-sample approximation via sampling without replacement

Theorem 4.2

Let $\bar{X}_n$ be the mean of a simple random sample of size n drawn without replacement from a finite population $\{a_1, \dots, a_N\}$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum(a_k - \mu)^2$. Then

$$E\bar{X}_n = \mu, \qquad \operatorname{Var} \bar{X}_n = \frac{\sigma^2}{n} \cdot \frac{N - n}{N - 1}.$$

Proof [Supplement]. Write $n\bar{X}_n = \sum_k Z_k a_k$ with inclusion indicators: $P(Z_k = 1) = n/N$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$ $(k \neq l)$, so $\operatorname{Var} Z_k = \frac{n}{N}(1 - \frac{n}{N})$, $\operatorname{Cov}(Z_k, Z_l) = -\frac{n(N-n)}{N^2(N-1)}$. With $\mu = 0$, $\sum_k a_k^2 = N\sigma^2$ and $\sum_{k \neq l} a_k a_l = -N\sigma^2$, so

$$\operatorname{Var}\Big(\sum Z_k a_k\Big) = N\sigma^2 \frac{n(N-n)}{N^2} + N\sigma^2 \frac{n(N-n)}{N^2(N-1)} = n\sigma^2 \frac{N-n}{N-1}.$$

■

$\frac{N-n}{N-1}$ is the **finite-population correction**. A CLT holds if $\min(n, N-n) \to \infty$ and $\max_k(a_k - \mu)^2/(\min(n, N-n)\sigma^2) \to 0$ (Hájek). For ranks, $\mu = \frac{N+1}{2}$, $\sigma^2 = \frac{N^2-1}{12}$:

$$E(W) = \frac{n(N+1)}{2}, \qquad \operatorname{Var}(W) = \frac{mn(N+1)}{12}, \qquad \frac{W - E(W)}{\sqrt{\operatorname{Var}(W)}} \xrightarrow{d} N(0,1).$$

With ties, use the mid-ranks as the population (mean unchanged):

$$\operatorname{Var}(W) = n^2 \cdot \frac{\sigma_R^2}{n} \cdot \frac{N-n}{N-1}, \qquad \sigma_R^2 = \text{population variance of the mid-ranks.} \tag{4.1}$$

Applied to the pooled observations it gives the large-sample permutation law of $\bar{Y}$, so permutation and normal-theory p-values agree for large samples.

4.2.2 Estimating the shift

In the shift model $U(\Delta_0) = \#\{(i,j) : Y_j - X_i > \Delta_0\}$ is the Mann–Whitney statistic of X and $Y - \Delta_0$; at the true Δ it has the null law, symmetric about $mn/2$.

Proposition 4.3

Let $D_{(1)} \leq \cdots \leq D_{(mn)}$ be the ordered differences $Y_j - X_i$, $k_l = Q_U(\alpha/2)$ and $k_u = mn - k_l + 1$. Then $P(D_{(k_l)} \leq \Delta < D_{(k_u)}) = P(k_l \leq U \leq k_u - 1) \geq 1 - \alpha$. The associated point estimator is the Hodges–Lehmann shift estimator $\hat{\Delta} = \operatorname{median}_{i,j}(Y_j - X_i)$.

Proof. $D_{(k_l)} \leq \Delta \iff U(\Delta) \leq mn - k_l$ and $\Delta < D_{(k_u)} \iff U(\Delta) \geq mn - k_u + 1$; use the symmetric null law of $U(\Delta)$. ■

In Example 4.1 the differences are -0.4, 0.9, 1.1, 1.8, 2.4, 3.1, so $\hat{\Delta} = 1.45$. For large samples $Q_U(\alpha) \approx \frac{mn}{2} - z_\alpha \sqrt{mn(N+1)/12}$.

Ranks as a probability transform. Under H_0, $U_i = F(Z_i)$ are i.i.d. uniform and $E\,U_{(k)} = k/(N+1)$: up to $N+1$, ranks are expected uniform order statistics. The rank-sum test is a permutation test after implicitly transforming F to the uniform; hence its null law is the same for every continuous F, and it needs $F = G$ (a common transform) under H_0.

Paired data. Apply Chapter 2 to $Y_i - X_i$.

4.3 Tests for scale

Assume a common center and $G(x) = F(x/\gamma)$. A scale change does not shift W, so use *scores* large in the tails:

- **Ansari–Bradley:** scores $1, 2, 3, \ldots$ from both ends inward; a small Y-score sum means Y is more dispersed.
- **Siegel–Tukey:** $1, 2, \ldots, N$ alternating from both ends, so the null law equals that of W.
- **Mood:** scores $(R - \frac{N+1}{2})^2$; a large Y-score sum means Y is more dispersed.

Each is a permutation test of a score sum, distribution-free under $F = G$. If the centers may differ:

- Under normality, $F = S_Y^2/S_X^2 \sim F_{n-1,m-1}$ when $\gamma = 1$; very sensitive to non-normality and outliers.
- $\mathrm{MAD}(Y)/\mathrm{MAD}(X)$, a robust alternative.
- Score statistics after centering each sample at its median.

Then $F = G$ is not the null, so permute the median-centered (mean-centered for the F ratio) observations; this is only *approximately* valid. Without centering, a location difference masquerades as a scale difference.

4.4 The Kolmogorov–Smirnov test

For general $H_0 : F = G$ against $F \neq G$, compare the empirical distribution functions $\hat{F}_m(x) = \frac{1}{m}\sum I(X_i \le x)$ and $\hat{G}_n$:

$$K = \sqrt{\frac{mn}{m+n}}\sup_x |\hat{F}_m(x) - \hat{G}_n(x)|.$$

The supremum is attained at an observation, so K depends only on the ranks and is distribution-free (continuous F). As $m, n \to \infty$, $P(K \le x) \to 1 - 2\sum_{j\ge 1}(-1)^{j-1}e^{-2j^2x^2}$ (Kolmogorov). KS detects any difference but is less powerful than the rank-sum test against a pure shift.

Why K depends only on ranks [Supplement]. Both EDFs are constant between consecutive pooled order statistics, so the supremum is a maximum over the N pooled points; at the kth point $\hat{F}_m, \hat{G}_n$ are the proportions of X's and Y's among the k smallest, determined by the ranks, which are uniform over $\binom{N}{m}$ splits under H_0. ■

4.5 Which test?

With $F = G$ under H_0 every permutation or rank test is exact. The Pitman ARE of rank-sum vs. t is $12\sigma^2(\int f^2)^2$, as in one sample:

F	uniform	normal	logistic	Laplace	Cauchy
ARE(Wilcoxon, t)	1	$3/\pi \approx 0.955$	$\pi^2/9 \approx 1.10$	1.5	∞

(Exponential shift family: 3.) The two-sample t-test is close to nominal for light tails and conservative for heavy-tailed or skewed populations. In small-sample power studies the permutation t-test leads for uniform data, the t-test for normal, the rank-sum test for Laplace-type tails and the difference-of-medians permutation test for Cauchy-type tails.

4.6 Unequal variances under the null

If H_0 is "equal locations" with different scales, no rank or permutation test is guaranteed to hold its level. Use Welch's t-test, a variance-stabilizing transformation, a studentized rank statistic (Fligner–Policello, Chung–Romano), or the bootstrap tests of Chapter 9.

4.7 Selected exercises

Motivation. The probability integral transform, why the Mann–Whitney count is not binomial, and the Ansari–Bradley null moments.

Exercise 4.6 (Probability integral transform)

Let $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} F$ with F continuous.
(a) Show that $U_i = F(X_i)$ are i.i.d. Uniform$(0,1)$.
(b) With $U_{(1)} < \cdots < U_{(N)}$ the order statistics, show that $E\,U_{(i)} = i/(N+1)$.

Solution.

(a)

Step 1. Fix $u \in (0,1)$ and let $x_u = \sup\{x : F(x) \le u\}$ (nonempty, bounded above since $F(-\infty) = 0$, $F(\infty) = 1$). $F(x) \le u$ for $x < x_u$ and $F(x) > u$ for $x > x_u$; continuity gives $F(x_u) = u$.

Step 2. So $\{F(X) \le u\} = \{X \le x_u\}$ and $P(F(X) \le u) = F(x_u) = u$.

Step 3. The U_i are functions of the independent X_i, hence independent.

(b)

Step 1. $U_{(i)} > u$ iff fewer than i of the U's are $\le u$, and $\#\{j : U_j \le u\} \sim B(N, u)$:

$$P(U_{(i)} > u) = \sum_{k=0}^{i-1} \binom{N}{k} u^k (1-u)^{N-k}.$$

Step 2. $EV = \int_0^\infty P(V > u)\,du$ for $V \ge 0$, and $\int_0^1 u^k(1-u)^{N-k}\,du = \frac{k!\,(N-k)!}{(N+1)!}$, so

$$E\,U_{(i)} = \sum_{k=0}^{i-1} \binom{N}{k} \frac{k!\,(N-k)!}{(N+1)!} = \sum_{k=0}^{i-1} \frac{1}{N+1} = \frac{i}{N+1}.$$

This justifies "Ranks as a probability transform" in Section 4.2.2. ■

Exercise 4.8 (Why $U(\Delta)$ is not binomial)

In the shift model, X_i and $Y_j - \Delta$ are independent with the same continuous distribution.
(a) Show that every indicator $I(Y_j - X_i > \Delta)$ is Bernoulli$(\frac{1}{2})$.
(b) Explain why $U(\Delta) = \sum_{i,j} I(Y_j - X_i > \Delta)$ is nevertheless not $B(mn, \frac{1}{2})$, as the analogous count was in the sign test. Compute $\operatorname{Var} U(\Delta)$ and compare it with $mn/4$.

Solution. Put $Y_j' = Y_j - \Delta$: $X_1, \dots, X_m, Y_1', \dots, Y_n'$ are i.i.d. continuous, $A_{ij} = I(Y_j' > X_i)$.

(a) $P(X_i = Y_j') = \int P(X_i = y)\,dF(y) = 0$, and by exchangeability $P(Y_j' > X_i) = P(X_i > Y_j') = \frac{1}{2}$.

(b)

Step 1. Indicators sharing an index are dependent. For $j \ne j'$,

$$P(A_{ij} = A_{ij'} = 1) = P(X_i < \min(Y_j', Y_{j'}')) = \frac{1}{3},$$

(each of three i.i.d. continuous variables is smallest with probability $\frac{1}{3}$), so $\operatorname{Cov}(A_{ij}, A_{ij'}) = \frac{1}{12}$; likewise $\operatorname{Cov}(A_{ij}, A_{i'j}) = P(\max(X_i, X_{i'}) < Y_j') - \frac{1}{4} = \frac{1}{12}$. Indicators with no common index are independent.

Step 2. With $mn(n-1)$ and $mn(m-1)$ ordered pairs of the two kinds,

$$\operatorname{Var} U(\Delta) = \frac{mn}{4} + \frac{mn(n-1)+mn(m-1)}{12} = \frac{mn(N+1)}{12}.$$

Step 3. This exceeds $mn/4$ whenever $N \geq 3$, so binomial quantiles would give too-short intervals; it equals $\operatorname{Var}(W)$ (Section 4.2.1), as $U(\Delta)$ has the null law of $W - n(n+1)/2$. In the sign test the indicators involve different observations; here each X_i meets every Y_j'. ■

Exercise 4.11 (Ansari–Bradley moments)

Let $N = m+n$ be even. The Ansari–Bradley test assigns the scores $1, 2, \ldots, N/2, N/2, \ldots, 2, 1$ to the pooled order positions, and AB is the sum of the scores of the Y's. Find $E(AB)$ and $\operatorname{Var}(AB)$ under $H_0 : F = G$.

Solution.

Step 1. Under H_0 the Y-positions are a uniform n-subset, so AB is n times the mean of a sample without replacement from the scores, each of $1, \ldots, N/2$ occurring twice.

Step 2. Population mean and mean square:

$$\mu = \frac{2}{N}\sum_{k=1}^{N/2} k = \frac{N+2}{4}, \qquad \frac{2}{N}\sum_{k=1}^{N/2} k^2 = \frac{(N+2)(N+1)}{12},$$

so $\sigma^2 = \frac{(N+2)(N+1)}{12} - \frac{(N+2)^2}{16} = \frac{(N+2)(N-2)}{48}$.

Step 3. By Theorem 4.2,

$$E(AB) = n\mu = \frac{n(N+2)}{4}, \qquad \operatorname{Var}(AB) = n^2 \cdot \frac{\sigma^2}{n} \cdot \frac{N-n}{N-1} = \frac{mn(N+2)(N-2)}{48(N-1)}.$$

Check: $N = 2$ gives $AB \equiv 1$, $E = 1$, Var $= 0$. ■